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The path of artificial intelligence from a fantastic idea to a scientific industry.

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Introduction

The topic of Artificial Intelligence is relevant in today's world due to the rapid development of the IT-sphere, new modern discoveries and the desire to improve, modernize and automate our lives. But do not think that Artificial Intelligence is directly related to the age of technology - the XXI century.

The birth of artificial intelligence

The first attempt to recreate the human mind was the creation of the first logical machine by Raimund Lullius about seven hundred years ago. Raimund's idea was to place pre-existing truths, concepts, and categories into concentric paper circles, and, by rotating these circles, to obtain new truths and knowledge. Though this idea was a failure, it was the first mention that reasoning could be done mechanically. [1 – p. 51]

Speaking of the birth of artificial intelligence, it is worth mentioning the achievement of Gottfried Leibniz, namely Leibniz's arithmetic machine. This device was a twelve-digit decimal device, capable of performing arithmetic operations, including multiplication and division. Over time, this device came to be called an arithmetic-logic device because a number of some logical operations were added to it. It was one of the first prime examples of a device capable of performing tasks designed for the human mind. [1 – p. 51-52]

The emergence of the Turing test

The main steps in the development of artificial intelligence as a scientific direction were made in the first half of the twentieth century, just after the first electronic computer. Then Alan Turing, a mathematician whose name is now directly associated with the development of artificial intelligence, in his scientific work put forward the following question: "Can machines think?" It was in this work that Turing formulated a reasoned proof of machines having AI, which soon became known as the Turing test.

Alan Turing presented his test in the scientific paper "Computing Machinery and Intelligence" in 1950. It has become fundamental in the field of artificial intelligence. The test is presented in the form of a "simulation game" in which three people participate: a man (A), a woman (B) and someone asking questions (C). The goal of the game for the questioner is to determine which of the other two participants in the game is a man (A) and who is a woman (B). «We now ask the question, "What will happen when a machine takes the part of A in this game?" Will the interrogator decide wrongly as often when the game is played like this as he does when the game is played between a man and a woman? These questions replace our original, "Can machines think?"». This question that has become central to Alan Turing's research. [2]

The Dortmund Seminar and its consequences

In 1956 a conference was held at Dartmouth to explore the concepts of "mechanization" of intelligence. It was at this conference that computer scientist John McCarthy coined the term "Artificial Intelligence". According to McCarthy, «Intelligence is the computational part of the ability to achieve goals in the world. Varying kinds and degrees of intelligence occur in people, many animals and some machines». [3 – p. 1]

Later participants in this conference would become prominent figures in the field of artificial intelligence. The most famous was Marvin Minsky, who in 1951 created the first network computer based on a neural network, a device built on the principle of organization and functioning of the nerve cells of a living organism. The pattern was called SNARC (Stochastic Neural Analog Reinforcement Calculator). It was the first self-training computer system simulating a network of 40 neurons. [4]

Thanks to the Dartmouth conference, since 1960 AI as an independent direction has spread around the world. The USSR, Japan and European countries joined the work. The period from 1945 to 1960 was the first step in the development of the new science. At that time the main directions of AI research emerged, the results of which gave rise to a great variety of ideas, generating both interdisciplinary connections and fundamentally new ideas. Experimental intelligent systems became the starting point for the development of such research areas as neural networks, game programs, maze search, machine translation, automatic theorem proving, visual pattern recognition and external influences.

Subsequently, subordinated to the trends of cheapening the hardware components of computers and increasing their power, the presented directions of AI research are transformed and intertwined with studies in philosophy, psychology, and cultural studies. Theories of "weak" and "strong" AI are emerging [5]. Writers and filmmakers in their works of fiction are trying to

show possible variants of interaction between humans and machines, touching on the issues of ethics, morality, and religion. AI enters the world market not only as a high-performance computer information system, but also as a global general cultural idea, which has generated both new hopes and new phobias in the minds of people.

Modern Artificial Intelligence

In the XXI century a lot of attention is paid to neural networks, as they are used in many areas of human life - medicine, engineering, physics, geology and even business. Thanks to neural networks, it is possible to solve problems of management, classification and forecasting. [6]

In the winter of 2016, Klaus Schwab, a German economist and founder of the World Economic Forum in Davos, said that artificial intelligence is the driving force behind the fourth industrial revolution. "We are at the beginning of a revolution that is fundamentally changing the way we live, work, and relate to one another. In its scale, scope and complexity, what I consider to be the fourth industrial revolution is unlike anything humankind has experienced before". [7 – p. 7]

Conclusion

To summarize, I would like to say that thanks to the computer revolution, AI technology has become our indispensable assistants. Now humanity is looking ahead, waiting for the creation of super machines. Although the consequences of the achievements of artificial intelligence are sometimes controversial, further development of this technology is inevitable.

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